



CS 465 Milestone One Guidelines and Rubric

Overview

You are currently working toward building a full stack web application for your client, Travlr Getaways. It is important to think about the overall system view of how all the pieces of the web app will come together to be a fully functioning website, which will include a customer-facing website, a database, and an admin single-page application that all need to work together.

Web applications are an inherently complicated undertaking because of the variety of complex technologies that must work together seamlessly and scale up to support thousands of connections in a short time frame. As part of your deliverables to the client, Travlr Getaways, you will need to prepare documentation to explain your process and the various components associated with the full stack web app.

It's best practice to document your plan for full stack development as you progress through a project. The Software Design Document (SDD) is an engineering document that contains technical specifications and some written descriptions and explanations. You will complete several components of the SDD in Milestone One.

Prompt

This milestone is intended for you to begin preparing the design documentation needed to deliver to your client, Travlr Getaways. Specifically, you will provide a high-level overview, a description of the design constraints, and a diagram that demonstrates the architecture of the web app. Keep your audience in mind and be sure to communicate clearly and concisely in client-facing documentation. Complete the following sections of the [Software Design Document](#) for your client:

Executive Summary: An executive summary is used to introduce the software design problem and present a solution. Be sure to provide the client with any critical information they must know in order to proceed with the process you are proposing. It will be essential to address the following in the Executive Summary section of the SDD:

- Describe the appropriate architecture of the web application based on your client's software requirements.
- Reference your use of the MEAN stack for development.
- Explain both the customer-facing side of the application and the administrator single-page application (SPA).

Design Constraints: Building an application has constraints. These constraints are things that provide boundaries or guardrails for the project and naturally come with limitations. Be sure to address the following in the Design Constraints section of the SDD:

- Identify the design constraints for developing the web-based Travlr Getaways application.
- Explain the implications of the design constraints on the application development.

System Architecture View: Component Diagram: A component diagram has already been created for you. Review the diagram provided and address the following in the System Architecture: Component Diagram section of the SDD:

- Describe the overall system architecture of the web application by referring to the component diagram provided in the SDD.

- Identify the significant components that will be used and their relationship to one another.

Please note: You will continue to add diagrams and complete additional sections of the Software Design Document throughout the course as you develop each aspect of the full stack web application.

What to Submit

Using the template provided, complete the following sections of the Software Design Document: Executive Summary, Design Constraints, and System Architecture View: Component Diagram. Submit the SDD with these sections completed to Brightspace.

Please note: You will continue to build upon this document throughout the course and deliver the final product as part of your project submission.

Milestone One Rubric

Criteria	Meets Expectations (100%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Executive Summary	Describes the appropriate architecture for the web application given software requirements	Shows progress toward meeting expectations, but with errors or omissions; areas for improvement may include discussion of client-side JavaScript frameworks responsible for rich functionality	Does not attempt criterion	15
Design Constraints	Identifies the design constraints associated with software requirements including an explanation of the implications of the identified constraints on application development	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	20
System Architecture View: Component Diagram	Describes the overall system architecture of the web application by referencing the component diagram provided and identifying the significant components and their relationship	Shows progress toward meeting expectations, but with errors or omissions; areas for improvement may include correct separation of components into physical tiers or comparison of three approaches for distributing functionality	Does not attempt criterion	40
Software Design Document	Accurately references the use of the MEAN stack for development throughout documentation	Shows progress toward meeting expectations, but with errors or omissions	Does not attempt criterion	15

Criteria	Meets Expectations (100%)	Partially Meets Expectations (70%)	Does Not Meet Expectations (0%)	Value
Clear Communication	Consistently and effectively communicates in an organized way to a specific audience	Shows progress toward meeting expectations, but communication is inconsistent or ineffective in a way that negatively impacts understanding	Shows no evidence of consistent, effective, or organized communication	10
Total:				100%